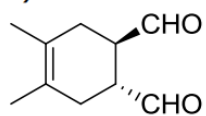


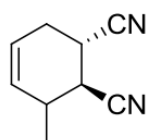
环状分子的非对应选择练习 (一)

1. Determine the starting materials used to form the following Diels-Alder products

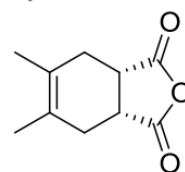
a)



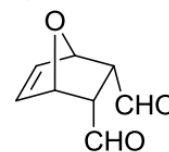
b)



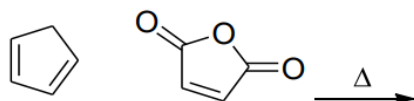
d)



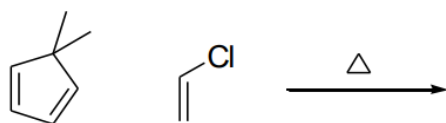
e)



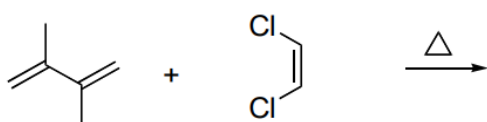
2. Predict the major product, if any, indicate stereochemistry where appropriate



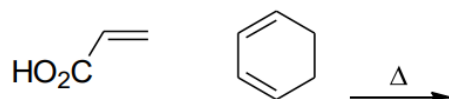
r)



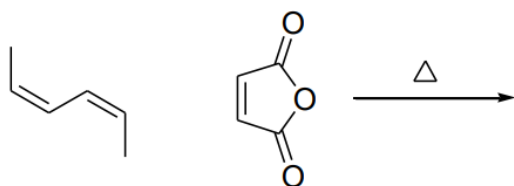
s)



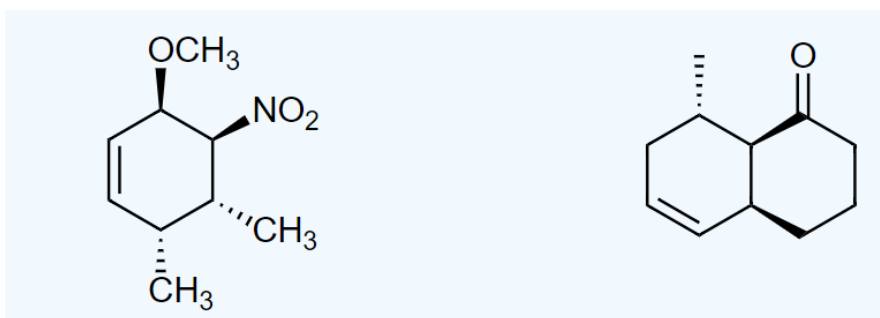
u)



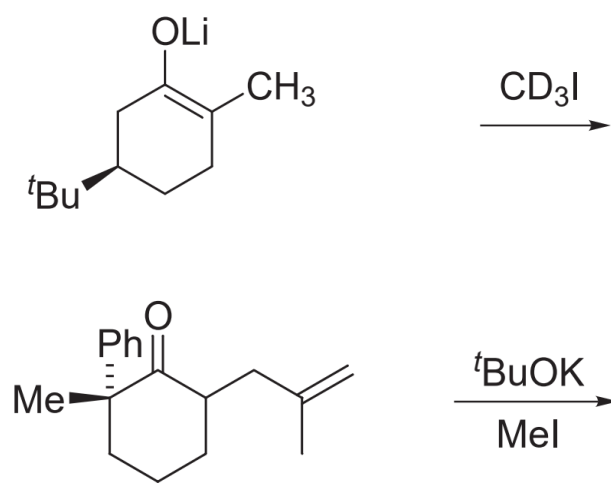
v)



3. Propose a Diels-Alder reaction that will yield each of the target molecules.

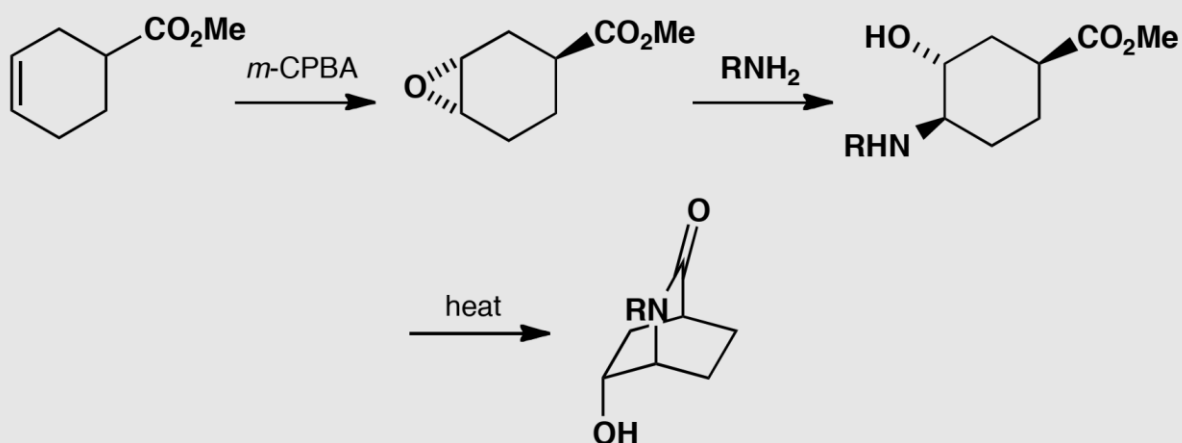


4. 写出以下产物的结构，并构象分析立体选择性。



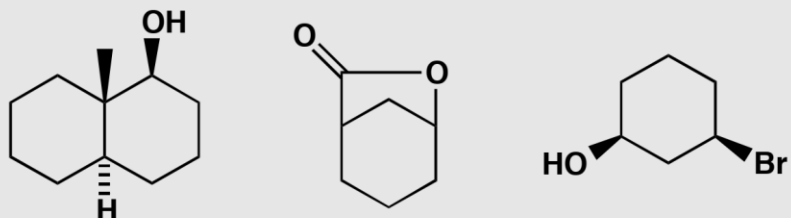
5.

Explain how the stereo- and regio-chemistry of these reactions are controlled. Why is the epoxidation only moderately diastereoselective, and why does the amine attack where it does?



6.

Draw conformational drawings for these compounds. State in each case why the substituents have the positions you give. To what extent could you confirm your predictions experimentally?



7.

Predict which products would be formed on opening these epoxides with a nucleophile, say cyanide ion.

